Lagerkvist, Amanda , and Jacek Smolicki , ed. Relational Technologies: In Search of the Self across Datafied Lifeworlds. New York: Bloomsbury Academic, 2025. Thinking Media. Thinking Media. Bloomsbury Collections. Web. 15 Jan. 2026. <http://dx.doi.org/10.5040/9798765118771>.

Accessed from: www.bloomsburycollections.com

Accessed on: Thu Jan 15 2026 16:04:44 Central European Standard Time

Copyright © Maria Rogg. Volume Editor's Part of the Work © Amanda Lagerkvist and Jacek Smolicki 2026. This chapter is published open access subject to a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International licence (CC BY-NC-ND 4.0, https://creativecommons.org/licenses/by-nc-nd/4.0/). You may re-use, distribute, and reproduce this work in any medium for non-commercial purposes, provided you give attribution to the copyright holder and the publisher and provide a link to the Creative Commons licence.

6

Relational interventions of Ogbon Gaia: 'Theory and practice of biohacking for designing co-existence 7,5 ECTS fall 2038 course code: 2IV168'

Maria Rogg

This chapter is a speculative fabulation of living a future of relationality through the second-person account of a university student in 2038. It presents a world governed by and striving for the principles of co-existence and care through educational and aesthetic means, following the strategy of Ogbon Gaia, to reimagine and mend a world that was reigned by and ultimately fell apart due to the hegemony of computational capitalism. It is also a lifelog, a written journal of the experiences and rewards of a university course on biohacking as an existential technique to intervene relationality, based on four modules that attend to the human as becoming capable of inventing alternatives to a violent, exploitative status quo. Following Haraway,[1] *speculative fabulation is engaged as an analytic and acts as a mode of worlding: through creating representations of alternative social relations, including ways of knowing, designing and living together;*[2] *ongoingness – the thick present of co-existence – is reconfigured. Storytelling, science fact, science fiction and feminisms are enmeshed in a collaborative political (thinking) practice to pattern possible worlds by re-figuring the semiotic-material movement of how the now informs the future. The present collaboration is a reiteration of the artistic and scholarly work of artists and designers who engage biohacking as a medium for new knowledges, imaginaries and collectivities that render a feminist ethic of response-ability. It draws on specific processes and strategies that transpired during participatory observations of various performances (see epilogue). Its normative standpoint*

[1] Donna Haraway, *Staying with the Trouble: Making Kin in the Chthulucene* (Duke University Press, 2016).

[2] Noura Howell, 'Emotion AI and Fabulation: Seeking Biopolitical Futures of Respectful Care' [Keynote speech], *Crafting Livable AI Futures*, Umeå (2023, 8 June).

unfolds via a narrative technique in which the action is driven by a character ascribed to the reader, an involuntary association that may be conceived as provocative. Intended as a rhetorical form of inquiry rather than a conclusive tool, the reader is affectively interwoven by creating access to experiential knowledges and embodied response.

Year two of the Master. You've heard the rumours. A new Guest Mover from Austria: a coder, performer, scholar and mother. The videos show them performing the aerial silks. A constricted body, a tree, the air and two strings of silk melt into a choreography of flight and support. The medium is conditioned by capacity and care. Limbs, movement, materials and gravity – extensions of ether, the element of potentiality.

The seminar begins with a historical recap. This is your take-away from it:

During computational capitalism, threats to existence on earth multiplied. Increasing scarcity of staples coincided with the progressive excess and abundance in richer parts of the world. While the disadvantaged died from planetary exploitation, the privileged boosted it. Competition for livelihood became fierce. By calculating ability, skill, ethnicity and race, profiling became the prime governmental strategy to regain control over automated terror, since robotic warfare had bred insurgent drone crimes and AI sabotage. Multiplied, morphed and disrupted on screens, one could no longer trust one's own eyes. A distorted picture had gained a life on its own. The privileged fell ill, drowning under an agony so all-encompassing that it became impossible to locate. States failed. When the EU collapsed, the UIK[3] suggested a global strategy to preserve existence on earth: *Ogbon Gaia*.

New initiatives were launched, radical at the time, with the purpose to recover life on a shattered earth. The criterion was cultivating an ethical response to all being. But what is the good life in a world whose fabric is torn? Do truth and justice compete? Who is responsible for which pattern and who carries the cuts?

In order to meet such concerns outside the solutionist space, the ruling knowledge paradigm was reversed. Policies to prioritize unknown potentials replaced the regime of certainty. This dominant, calculative way of reasoning had been identified as complicit with the prevalent loop of alienation and violence. Methodologies to linger and learn from present cracks became subsidized. And while processes blossomed, a lot of people lost their jobs. Gradually, everything that had been taken for granted became a matter of inquiry. Then, much of what had been certain had to be unlearned. Unrooting the foundations of our knowing was a big blow. It drove a lot of people mad. After having lost their homes, jobs and status,

[3] Union of Indigenous Knowledges.

everything they believed in was taken from them. Alienation prevailed, but in a different way. Artistic knowledges gained influence as an intervention to provide orientation by making co-existential balance purposeful. If the goal was living well, in response to all other beings, an understanding of this experience had to be gained. The arts promised several advantages in this regard: Through art we can make *sense* of the immeasurable – of uncertainty, disorientation and loss. Art is an ancient technique to develop perceptive sensitivities for the human condition in a particular moment. Ultimately, it was believed that once we perceive attentively and become attuned to our experience in relation to the world, we are better equipped to choose what we need to maintain this equilibrium. And indeed, it showed that those who had a place in one of the many projects adopted quicker to enforced uncertainty. Involvement provided an invisible infrastructure of support. Art irritates, because it confronts us with our limits. By planting a seed of imagination – it could be different – art becomes an existential rite to initiate alternatives. A different story is a venture. Risky, but, after all, an invitation to get involved: to interrupt, interrogate and infuse the other by affirming both worldliness and the individual reality of ourselves. That way, our projects shape each other and we relate. An absolute future doesn't exist. Futures are made.

Thus, the *New New School.*[4]
Thus, the course.

You are further reminded that biohacking had emerged long ago but had taken a relational turn with Ogbon Gaia, just like everything else. Historically, it actualized the body-machine metaphor, 'serving' people who incorporated technology to enhance their abilities and boost their independence.

Back then, independence was a cruel standard. Profoundly entangled with privilege, its virtue turned into ideology dressed up in deed. People did all sorts of harm in the name of independence. As a condition for both control and liberation, independence is a double-sided sward. And it continues to be. Under Ogbon Ori, independence is supposedly maintained embedded into the relations that co-constitute it. Critics claim, however, that that's an ideal

[4] The department's overall objective is to unite research and education around sustainable societal development (and the readjustment such development requires) via a radical rethinking of the individual-society-environment relationship. By challenging conventions in established areas of knowledge, by working transdisciplinarily and by uniting interventionist practices with advanced epistemological and methodological perspectives, research and education in the programme shall contribute a strategic alignment of innovativeness and maintenance for liveable futures in harmony with planetary existential boundaries.

which favours self-harm for those excluded from it. As usual, questions of being compete against power imbalances.

Today, the purpose of biohacking essentially lies in deconstructing the Human ideal that motivated the abuse of independence, by crafting, sensing and testing co-existential experiences, often, but not always mediated technologically.

Co-existential modes of knowing and doing conceive of human beings as differently situated, unique and inexhaustible individuals that nonetheless share a range of qualities, needs and entanglements. Human existence is defined by vulnerability and finitude due to its embodiment, a singular-plural, intracorporeal mode of being that unites rational and intuitive knowledges in ways that exceed final comprehension because it is always already related to others, tools and the inhabited world. Thus, it is subjected to contradictions, inconsistencies and experiences of uncertainty; it is fallible. Yet, human existence is conditioned to make choices. Being human means both having and being a body before *other bodies* in shared vulnerability. Therefore, it means being obliged to act on the suffering of others. It means being ethically respons(e)able. Human subjectivity is an emergent and ambiguous relational process that counters the fabrication of *men* as independent, universal and absolute. And yet the fiction of an autonomous, unwounded and certain Human, separate and superior to everything and everybody lacking these attributes got solidified algorithmically.

From the beginning a philosophical construction, this tale of the Human, the 'liberal humanist subject', had proved profitable for a small group of the global population who gained wealth, freedom and power by nurturing the idea of their exceptionalism. During late modernity, automated architectures were built to cement it. Known as *biometricalization*, living bodies were abstracted into discrete data identities to calibrate their norm coherence and utility. Biometricalization existentially endangered the othered population, non-white, non-masculinist, non-heterosexual, non-ableist subjects and induced severe harm to the environment. The norm enforcement found an expression in biohacking as a cultural technique to discipline the self and transcend ageing, disease and eventually death by wiring embedded and embodied technology to self-monitoring and quantification practices. In battle against vulnerability, the movement's mantra 'know thyself' called for enhancing a metric non-sense of the self in pursuit of autonomy while spiritually perpetuating captivity. 'Transcape' spread as a neologism for the escapism involved in seeking revelation by dissecting the self as a database. Ultimately, biohacking personified the Cartesian, masculinist-individualist ethos that governed (digital) existence and gave it its techno-utopian face.

The recap also brings to the fore that after the launch of Ogbon Gaia, biohacking was repurposed as a co-existential media technique. It provided a methodology and practical repertory to implement the relational paradigm by refiguring the human (imaginary) as entangled and co-emerging, not only with digital technology but also with others, nature and things. It's a layered technique. The foundation is the reconceptualization of the body from being seen as an object and servant towards being a vehicle that middles between self and other. Adopting a hacker lingo, the body could be described as the ultimate blackbox: an impenetrable system whose inputs and outputs are known but not fully understood, a token for a complex, productive, volatile and substantial, yet inextricable force. The metaphor facilitates to reimagine the body as a medium of memory, trauma and affect which equips us with a story of our own and of society. Our anxieties speak of wounds inflicted upon each other and our environment. Against salvation, our bodies remind us to admit crisis as a present state which requires to incorporate experiences of struggle, grievance, responsibility and hope in search for practices that make action meaningful. Just as systems and futures, bodies are made. They aren't isolated but enmeshed with the materials, textures and structures of the world; the belief systems and values, limits and *bodies* that societies choose as normative. In and through them, history is made. Thus, it is through our bodies that we learn to live otherwise. To sense, to feel, to know and to do otherwise. The body as the nucleus of this practical wisdom is the starting point of biohacking as we think of it today.

Biohacking is a controversial term because hacking denotes a disruptive form of resistance and remodelling of a system, a practice that breaks the walls of the box. As a hack of the human body, it strictly needs to be understood historically, as a *political* term agitating against the tokens of digital existence: it was not so much the body subject that was hacked, but the digitally decoupled, superior and absolute human imaginary the body had been enciphered with for so long. Biohacking became a movement and practice to reappropriate the existential body as a relational, ethically responsible and politically situated body-self, a relational autonomy that was bound to perish under the digitized universalistic narrative. If we had continued to see humans as separate, we hadn't been able to care for our vital habitat. If we had continued to see humans as abstract, we had kept secluding our essence in self-destructive bliss. Finally, if we had continued to violate our difference, we would altogether have seized to exist. These toxic fabrications of 'bio', of life, had to be hacked, to envision, experience, and practice how the implication of human beings in networks of care for other humans and non-humans actually work.

The course introduction stresses that biohacking operates in layers. When an awareness for co-existential qualities and needs has been established, hacking the body is surpassed for layer two: hacking biopolitics, the taxonomies, tools and techniques used to uphold the status quo. System hacks mainly consist of thwarting essentialisms and other institutions of binaries. Remodelling them, the effective design of co-existence, transpires in layer three: once the effect mechanisms of the past are internalized, the third layer of biohacking embodies iterations of crafting, probing and testing new experiences as an ethical practice of the co-existential paradigm. To remain careful, a reflective cross-layer is integrated. It serves both as a corrective and a coping tool to relieve the ambiguous trauma caused by the suffering of the past and the nonetheless poignant grief about its loss.

You are re-familiarized with the fact that the methodological and technical inventory of biohacking is as wide as its objective, the remodelling and cultivation of the human imaginary towards co-existence. In pursuit of that, technology provides a medium for relational, resistive and restorative modes. It is equally used to hack systems of oppression, as it is engaged as a potentiality to prompt and curate experiences which mediate co-existential values and habits. In fact, gait and sound recognition recently became reintroduced to observe and trace animals in order to identify biodiversity loss and protect vulnerable species.[5] However, the use of technology doesn't define biohacking. Many would argue that it's a technique defined by its purpose, to mediate and practice experiences of the self as an ethical agent who exists in perpetual becoming with others and the world.

The course you have taken up contains three modules to practice the three layers of biohacking which integrate the cross-layer. Each module consists of performing, as in en-acting, a layer-specific biohack. Afterwards students are expected to evaluate its purpose and means, and present a critical question that qualifies the next iteration by reflecting ethical ambiguity. You had previously taken classes in performance studies and the methodological choice was clear: the pedagogy of performance integrates coping mechanisms with strategies and tactics to investigate and intervene in reality, because what happens is co-shaped by how things feel. It's a circular pedagogy that combines sensing, coping, learning and making. The integration of being a test ground and mood barometer attunes the four layers of biohacking. Further, performance is a relational remake of the present *in situ*. Emplaced

[5] In 2037 alone, the UIK reported 30888 poaching delinquencies that corrupt biometric conservation efforts for precision hunting: https://mitpress.mit.edu/9780262048750/gaias-web/.

in a particular moment of time, its essential components are a performer and a recipient who can switch roles at any time. Without this relation a performance seizes to exist. It produces a space that is designed together, where each sound, movement, breath and matter matters in the reciprocal mediation of an otherwise. This knowledge generation is affective, relational and emplaced. It's a site of practical judgement. Here, ethical negotiation takes place. Reality is constructed as a reflection and negotiation of the present. It's actuality and future at once: future-in-the-making.

Module 1: Hacking the human

You enter a spacious and light room furnished with circularly arranged yoga mats on the floor. The windows are big; sunlight shines through. You and your classmates are instructed to take your shoes off and find a comfortable seat. You are introduced to the scenario:

> In this module, you contribute to »the Library of Toxicities«; an online support group which dates back to 2022 and provides advice and resources on cleansing the body from material and immaterial toxicities. You will collect, explore and discuss toxicities. Based on physical and reflective exercises, you will also generate new co-existential systems of classification.

The mover (FKA teacher) starts the exercise with a data collection of toxicities. They explain:

> Today, we are just collecting data, there is no valuation happening of that data.

Each student gets to complain for a minute or two. Their neighbour transcribes this 'data'; the rest listens actively. Displacement, alienation, doubt, mental anguish, ageism, exhaustion and objectification are expressed as symptoms of disintegration experienced since the turn. In the past, people suffered from enforced lack of (self-)care, isolation, toxic relationships, exhaustion, violence, mental health, racism and toxic work environments. You're curious about the relation between today's complaints and the toxicities back then. Both describe a lack of autonomy, yet under different regimes. Then, the group is asked to reflect non-judgmentally about varying proximities to toxicities among the group members, and complicities in generating them.

> This is important to maintain agency and autonomy without falling into the trap of isolating oneself in face of these asymmetries.

The class continues with a reflection of the sensorial experience of the exercise. Several students describe embodied shifts: a closed throat, restricted breath, shifts in temperature and (dis)comfort.

> The body is a witness that stores information in relation to trauma and toxicities. Let's support it by teaching it something new.

During the break you have a chat with Kim, the classmate whose complaint you recorded. You could recognize yourself in her account of relationality which actually isolates. Kim shared that her care work simply takes too much energy from her to remain fully engaged. Her own disintegration results in a withdrawal from the world. She's neither here nor there, but everywhere. And yet, the sense of worthlessness is hers alone. A strange dilemma of self-dissolution and re-individuation at once.

You thought it was brave of Kim to share this experience in the class. After all, the New School is an authority.

Anyhow, Kim and you agree that the exercise generated a sense of belonging and solidarity by giving each person a space in unfolding collective agency. The temporary suspension of judgement allowed grievances, including guilt and shame when revealing struggles or confronting unjust privileges. Complaining without moralizing, Kim believed, increased tolerance for care as a matter of capacity. We're human; we stumble, we fall, but we are doing what we can. Still.

Once everyone is back, the transcriptions are silently piled up in the middle of the room, a solemn gesture announcing the next passage. You sit down in a circle, facing each other and the trash pile of collected toxicities. A soundscape of waves and birds fills the room. Sitting in a sublime lotus position, they invite you to release tensions by paying attention to the breath and turning your gaze inwards. The group sits close to each other, yet there is enough space to move without touching each other. Collectively, you do a standing exercise, facing each other's backs. Through the physical proximity and aligned movements, you experience closeness with the others while feeling intimately connected with your embodied self. You sit down and they introduce the next step, a physical trauma release exercise. You are told that the body usually starts to tremble and introduced to a resting pose that you may return to at any time. You put the soles of your feet together, knees pointing towards the sides, and lift your hips. After a while, your legs effortlessly start to move up and down, like wings. Stimulated by the

involuntary movement and the ambient sounds you begin to picture yourself as a frog. You stifle a laugh. That was unexpected. Then you surrender.

You gaze up into the air and sense movements next to you, yet nothing concrete. You're focused on your sensations and the physical effort needed to maintain the position. After three iterations, the group meets in the resting pose. Then everyone sits up and re-enters the shared place. Your classmates look dreamy, sighs of release travel the room. They start the reflection, by placing a beautiful plate of nuts and fruits in the middle of the circle.

I always get hungry when I do this. What did you experience?

You share your sensational intimacy with the animal realm. They explain that this makes perfect sense, due to the activation of the Psoas muscle, an ancient, primal muscle. Your classmates share feelings of tiredness, relief, and anxiety over losing control: 'I feel kind of vulnerable in a way that I can't say. It is hard to articulate vulnerability. I think it is connected to the loss of control, like, wow, there's something that I didn't know about me.'

They nod. 'Today, we taught our body something new. It's training. The trembling is an expression of existing trauma and can seem very alien. It can feel destabilizing to surrender to it. But the body is speaking. It has its own agency.'

The final part of the performance consists of making new taxonomies by returning to the transcriptions and shouting out categories of toxicities that we think of in relation to our complaints. The contributions of the class are added to the Library of Toxicities. You sign up for it.

Then, they turn to the students and ask: 'In which way did this performance enact biohacking as a co-existential practice? What was its purpose and which techniques were used?'

Kim answers:

> I believe, that the purpose of this performance was to manifest our relational being and autonomy. We performed layer one which always requires an emphasis of the cross-layer. Co-existence still *feels* alien and demands care and support strategies that facilitate the alleviation of insecurity and Angst. We embodied the limits of rational knowledge by practicing subjective, sensuous, and caring ways of knowing that welcome rather than battle uncertainty. This training in existential knowing reveals aspects of the relational self: a self that is integer *and* permeable; autonomous but not isolated. Being, that is acknowledging that we are always already relational, became tangible through modes of intercorporeality and shared vulnerability. Based on a judgemental-free

zone a collective emerged despite conflicting individual positionalities. The trust created embedded the authentic expression of struggle, need, joy, and pain, a co-existential dialogue.

Very well, and your question?

Kim sighs:

I feel my body yearning for closure. Where is silence and integrity in all our embeddedness? Also, I doubt that our diverse ways of knowing can become socially scaled and yet remain free of reductivism. How can our existential knowing inform the rational and vice versa, so that we can seize them politically, as existential measures?

Module 2: Hacking the god trick

You enter a workshop. It's warm. A thin layer of dust covers windows, test tubes and drawers from the past. And yet the place is vibrant; a beehive of cables and electric humming birds, filled with used things and things in use: printers, projectors, panels, microwaves, drillers, hard drives, stereos. The class sits down around a big blue table silently speaking of the upcoming activity. It presents zines and academic journals on gender identity; a collection of small colourful objects that turn out to be miniature genital models; a 3D printer; and a razor-sharp tool, a forceps with a ruler attached.

Then, from nowhere, a willful artificial voice claims, 'Bodies Are Not Born, They Are Made.' A picture is thrown against the wall: The tool, followed by rising sea level that is contaminated by plastics, pills and tubes. A title appears 'Alien Genital Scanning for all Unruly Bodies.' A video tutorial introduces you to the module's outset:

During computational capitalism, intersexuality was systematically removed at birth. At the same time, the rising absorption of plasticizers was changing prenatal levels of androgen, a pollution-based hormonal disruption that led to an increase of physiological intersex features. Classified and feared as reproductive toxicity, this mutation was carefully observed, measured by means of the "AGD", the distance between the anus and genital. The AGD is a sexually dimorphic measure; males used to exhibit a 50 – 100% longer distance than females did. However, since hormonal disruptions through plasticizers male AGD's had been steadily decreasing.

The voice reappears:

> In the wake of hormonal disruption, it is clear that our bodies are not as fixed and recalcitrant as we want them to be. The grain proximity tells us something about where our anatomical existence wants to go. Is it possible that we are all approaching intersex? Isn't it time to redefine genital aesthetics – Isn't it time for new genital taxonomies?

Shortly after, the mover enters the room and introduces you to your task:

> The purpose of this exercise is for participants to biometrically capture, model and 3D scan their genitals and donate this data to an anonymous, crowdsourced database, »The Archive of Unruly Bodies«. The archive is a counter-hegemonic intervention that recognizes bodies as unfixed and changeable, by means of conducting a queer population study based on the data that participants donate. Instead of their gender assignment at birth, participants register their chosen gender identity. That way participants become the primary deciders over their normativity.

You go to a separate room to take your genital scan. It does not work. Your genital is not recognized. Your genital does not work. You try again. It still won't work. It's hot and the walls seem sticky and tight. A sudden revulsion takes hold of you. Apparently, you are already 'contaminated' and this is how you find out. It's disgusting. To hell with entanglement. The naked failure trapped here is you, you alone. You are hesitant and wonder: *How is multiplying alienation supposed to help us relate anyway? What are we actually hacking and for what end? What a bullshit task.*

You breathe. This is part of the process. Co-existence still *feels* alien and demands care and support strategies that facilitate the alleviation of insecurity and Angst. You agree to try once again, but with a different device. You borrow their smartphone and this time it works. Your bioscan appears.

Each scan is transferred to an individual computer that contains the required software. Some students are already familiar with it and can support the rest of the group. As a means of facilitation, their editing process is projected on the walls. You observe how their genitals gradually take shape. There is no sense of hesitation or shame in the room. Emotional intensity has decreased since the turn. Temper is held low in consideration of its distributed effects. It's a matter of decency. Yet there is more to it, as you begin to understand when you start modelling.

The technology absorbs you. You're swallowed by designing in 3D. It requires a particular gaze that makes you lose sight of the intimacy processed.

The more you model your genital, the more you're detached from it. You could be editing anything. After hours, you donate your scan. #278.

The day after, the class meets again. The printer is silent, but its fabulations remain. A collective pelvis, ten queer, silvery genitals placed upon one gluey tray. Yet, your print looks suspiciously alike the print placed next to it. Your fellow student observes you, and says:

'A glitch in the database. I think you mistakenly modelled my scan. It's interesting, 3D-printing re-materializes the body and the body loses more quality each time.' Then, reflection starts again and this time you yourself answer:

> I think we performed layer two of biohacking, a system hack, by appropriating and reenacting biometricalization to remodel and queer the datafied self. The essential logics and properties of biometricalization are the allegedly objective metrification of life though normative databases as an undisputed authority. By meshing, recombining and repurposing traces of the digital and the embodied self, biometrics were both mimicked and debunked for an operation of becoming. Rather than solid function, a fluid, multiple and relational process was materialized. Thereby, we in fact hacked gender which might have rendered feelings of intrusion and loss.

'Right', they respond,

> but let me add some explanations: As suggested, the module put forth a biometric counternarrative, based on two colliding limit situations: the enforced removal of intersexuality which caused severe harm to people with ambiguous gender identities; and, the agency of pollution in deconstructing the sex binary. As identified correctly, the performance addressed how biopolitical metrics of computational capitalism suppressed volatile embodiment and existential entanglement with, for example, technology. Here's what I'd like to add: Patriarchal normativity was also hacked by reframing markers of objectivity such as science, technology and materiality. Performing science can thus be seen as a method towards the active cultivation of co-existence, *the third level of biohacking*. On the one hand, it highlights scientific limits: How do you archive a body that is constantly changing? On the other, it encourages to explore alternatives, through Other scientific fictions. Finally, the intention behind students' learning to master biometric technology was to mobilize critical reflection.

'I thought of that, too', you reply. 'The genital modelling set off a disaffection resulting from the datafication of bodies. At the same time, it made our bodily volatility tangible.' Remembering your critical task, you add: 'And it made we wonder: Are values such as integrity amendable when making an image of an imagined reality the reality of art/an art reality/a reality through art? How do we avoid the establishment of new normativities, let's say, enforced transparency, in the redesign of existing norms? Where lies the authority over distributed bodies, if we ourselves are lost?'

Module 3: Designing co-existence

You enter a theatre. The walls are painted in dark red and the light is dimmed. You sit down in one of the front rows to have a better view. A slender figure stands in front of the movie screen, holding a device. Pale, with blueish hair, like ice. Thick blue lines repattern their face. And yet, stillness defines this face. Is it a mask?

Then, from nowhere, a defeating scream of pain pervades everything: the room, your body, their face. The person lifts their wrist and, abruptly, the sound disappears. Behind them, a sonogram fills the screen. In unwavering self-evidence, shades of grey pump up and down. On the lower right, a visual frame highlights impulses in colour. Red, orange, yellow, blue transgress like an old-school weather map. On the left, a fragile upper body shows. Above the heart, the skin is curved. What's under it?

The picture changes: 'Please take up the headset stored under your seat and start the device.'

A notebook appears. I.I. 2017, it says on top. The person remains still, looking at you. Then a child's voice enters your ears:

> What you see is my diary, an existential medium. I was born in 1982, somewhere along the autism spectrum. This wasn't seen, nor heard. People were busy trying to fix me, or getting angry as it wouldn't work. I connect better with the non-human world. Its communication is more straightforward. If a cat has mixed feelings, I can tell. Thus, I did not want to eat cat, or any other animal. My people thought that not eating meat meant not eating anything. I was called anorectic and force-fed with pork. That was one of the situations where I would lose my voice. My capacity to speak. To speak up. Back then they thought that not communicating verbally meant not having anything to say. Once, a tumour that required surgery was found in my leg. The doctor was

angry, because of my piercings. Never before had I experienced such pain. Yet, I/he/we couldn't tell. I silently fainted. He yelled. I would have given you painkillers, but you didn't scream!' Years later I would find out about Selective Mutism: a sudden, yet total incapacity to speak when experiencing emotional distress amongst neurodiverse people.

I hacked the technology of my diary. Digitalization had taken flight and instead of piercings people implanted microchips under their skin. While they used it to instruct their front doors, I gained a voice. If I'd hold a reading device close to an NFC implant, the data stored on the chip could be translated and displayed via a magnetic field, both visually and sonically.

Technically, the chips were crap. Data had to be stored locally on the hardware. The benefit of this limitation was that no data could be shared without its carrier's consent. The need for physical proximity was also beneficial to stay in control of one's data. However, convenience and lack of imagination led people to agree to whatever terms of condition offered to them. I used my implants to communicate my feelings when language would fail. Extending my body provided a possibility to set limits that people with a normative body assume as natural. By borrowing my brother's scream, I reclaimed my vulnerability. By sharing my heartbeat, I sought care.

NFC implants belong to the archives. The technology is obsolete. Today, there are less invasive means to diversify communication. Now, you will invent one of them. Follow me.

Smoothly, they – the tutor for this session – climb down the scene and walk towards a portal you hadn't noticed. Cosmic percussions fill your headphones. Sprawling, wafting beats ascend to cracking ice. You picture the landscape of a mountain range. Without pressure to escalate, the galactical ambient creates tension. You wonder if everyone hears the same. The collective seems to share an orientation, yet ultimately, you cannot tell. It is interesting to watch: a swarm of elements embodying a rhythm on their own.

The flock enters a hall divided into craft corners: A kitchen, a sound studio, a textile plant and a lab. Intuitively you walk towards a line-up of sewing bords, furnished with components to make moderate interaction designs and a catalogue of the biomaterials grown in the lab.

The tutor explains where you and your course companions are now:

You're at the station for soft technology and bio-textiles. Here we ask: Can Wearables change the ways our bodies relate? Your task is to design a non-extractionist and non-intrusive wearable, based on seizing one

of your own limitations for a remediation of voice. You will manually craft your own wearable electronic. Basically, that is an integration of technology and apparel to create a loop through which electricity can travel, communicate and interact. Through conductive thread and a sensor that absorbs the input, your biodata is connected to a micro-computer that processes and translates the information into a different output. That way, you can expand the sensory surfaces of your body and interact with the environment in new ways. I recommend that you start by deciding upon a desired outcome. Which relation are you envisioning? How would the interaction feel? You can affect this sense of interaction by deciding on the input and the output of your relational design. Former is the living information that your textile can absorb, reflect, measure or generate. For example, your breathing pressure, bodily fluids, shifting temperature, or simply touch. Latter is a remediated response of your choice.

While your gaze registers the available fibres, actuators and transmitters, your fingers hover over the conserved textures in the catalogue. Each is captioned with a description of the qualities, production and application areas of the biomaterial. While the production of electronics is quite dubious since the turn, all legal fabrics are biomaterials, living matter grown from renewable biomass, such as algae, mycelium, kombucha or crystals. They are transient entities that breathe and transform their colour, shape and texture in environmental response. Depending on the quality of contact, their resilience, elasticity, durability and feedback to added attributes, say heat-sensitivity and conductivity, vary. Each fabric is unique. Once their life circle ends, they are disposable.

Ffff! You exhale in pain.
One tilt!
At least, you know what to do now. A refigured hacker's diary. A relational lifelog …
You start to write.

Lately, I've been thinking more about the making of knowledge as a political project. I read early classics whose visions for co-existence were marginalized before the collapse. Their sabotage and revival speak of knowledge production as a means to sustain or counter oppression, and hence, as a contested matter of care. Storytelling has always been a way of producing and reproducing knowledge because stories explain the world to us. What stories we tell and how we tell them acts. In that way, stories

have a design agency. A transition text by Ursula K. Le Guin illustrates how stories (re)shape the world. By projecting a world on their own, they become living fabulations. »The Carrier Bag Theory of Fiction« describes how the main epos of the past constructed The Human as a hunter who survives because He subordinates. By telling a different story, a story that replaces His spear for their carrier bags that are used to gather and spread seeds of/for/to life, Le Guin twists the dominant narrative towards a life story within relations between generations, sphere, and species. She wrote this in 1986, 'with a certain feeling of urgency'. Today's total conflation of life and urgency suggest that our living stories can't be told often enough. Accordingly, my design challenge goes as follows: How can I design a non-intrusive medium that spins our webs of care through living stories?

Currently, my life is marked by back aches, because I carry a child. My design is a wearable recipient of life stories, triggered by a simulation of my own story in response to a shoulder movement caused by tensions experienced throughout my pregnancy. Pregnancy has always been a situation at the limit and it is no different in our age. As a prism for the profound ambiguity of existence it demands recognition for life that is both fluent and frail. Thus, pregnancy is as much a revelation of relationality as its guardian. It presents us with our multitude while insisting to draw the line – to live limited by our vulnerability. Reminding us that being plural throws us into precarious situations that entanglements cause, pregnancy is a quest to face our eternal uncertainty and thread a texture between evoking and debunking essences to negotiate and affirm the world-making capacities of human collectives. A quest to seek knowledges from within the womb.

The circuit is sewed into the shoulder strap that connects an LED light (the output) to a stretch sensor activated by the shoulder movement (the input). Thin films of Agar bioplastic are casted over the circuit, resembling biodegradable plastic foil. Its transparency presents my body, technology and the fabrics as an assembly, whereby the recipient ultimately tells a life story on its own. When I move, the bag lights up and opens its 'mouth', the container. This sensory extension sends a signal that people might attend to. They might ask me what this is all about. I'll tell them my story, and then I'll tell them yours. I'll ask about their situation, their knowing, their needs. We might render each other capable of responding carefully. We might find a language for our shared condition of relating differently.

The last module is over. This time, you are asked to reflect in silence. You look at your text and realize that it sums up the living story of your participation in the course.

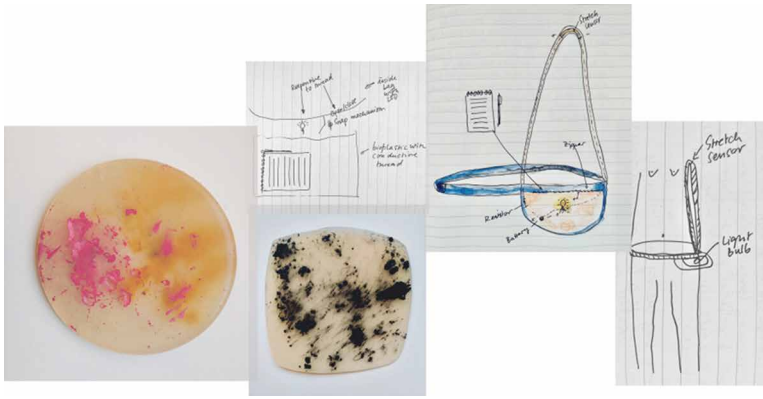


Figure 6.1 Wearable recipient, 2022, Maria Rogg.

The Greek word *historia* means 'inquiry, the act of seeking knowledge, as well as the knowledge that results from inquiry'. Starting from a certain history of Human entanglements with technology, your story became an artefact to embody his-story otherwise: a materialization of our co-existence with technology that makes explicit our human condition – our capacity to narrate, imagine and enact alternatives.

Thus, *Module 3*: *Designing co-existence* aligned all layers of biohacking, providing you with the means for relational interventions. The intervention integrates each module: (1) the human imaginary is reoriented towards relationality. A human being that is unique and uniquely situated enters the picture, who experiences vulnerability subjectively, while sharing it through the quest of responsibility. (2) Stories are unveiled as an analytic, a knowledge-making practice for real fiction. That is based on (3) twisting the lifelog. Once a record of mistaking existence as a Big Dataset, the lifelog transforms into a technology to increase knowledge about co-existential stories that cultivate care. X. This provides the possibility of a living response – a response-able process towards a situated ethics of care.

But: How do you know that the response triggered is right, or 'ethical'? Are the waters of care politics ever safe?

You remember a passage of another transition text, written by Amanda Lagerkvist:

> Beyond myths about a sovereign and all-knowing subjectivity, the exister is neither omniscient, nor 'the measure of all things.' The exister is mortal. She is a struggling, suffering, and relational human being. Her

> intermediary position as a conscious (yet often clueless) embodied being provides for her the only known 'entry point' to navigate and craft the world into which she is thrown. Conceiving of the existential this way both reclaims and resituates subjectivity as it at once centers on and destabilizes 'the human.'[6]

You contemplate.
What does it mean to be human in the age of Ogbon Gaia?
You aren't certain. But you know ways to learn.

Epilogue

This story emerged from an in-depth empirical and hermeneutical exploration of biohacking as an existential practice, based on interviews and participatory observations. In particular, it draws on specific processes and strategies, experiences and encounters that transpired during the observant participation in various performances. Artists Mary Maggic and Idun Isdrake and designer Giulia Tomasello must be mentioned in particular. Their practice, philosophy and politics are the patterns that solicited response. Specifically, Module 1 is inspired by the Toxic Consultancy workshop hosted by Mary Maggic and produced with the Artist Project Group at Galerie Elisabeth and Klaus Thoman in Vienna, 2022. Module 2 is based on the workshop Genital (*) Panic: For all Disobedient Bodies, by Mary Maggic at Kersnikova Institute, 2022. Module 3 combines scenes from Idun Isdrake's work Radical Imagination and Giulia Tomasello's online course Coded Biophilia taught at the School of Machines, Making and Make-Believe in Berlin, 2021. The collaboration also involves the entanglement of the creators' work with academic scholars, concepts and fields equally committed to care-ful response. As a genre modality, almost all in-text citations are sacrificed, but existential media studies, feminist new materialisms and critical data studies are the conductive threads for fabulating this knit.

Bibliography

Ajana, Btihaj. 'Recombinant Identities: Biometrics and Narrative Bioethics.' *Journal of Bioethical Inquiry* 7, no. 2 (2010): 237–58. https://doi.org/10.1007/s11673-010-9228-4.

[6] Amanda Lagerkvist, 'Existential Media: Toward a Theorization of Digital Thrownness', *New Media and Society* 19, no. 1 (2017): 101. https://doi.org/10.1177/1461444816649921.

Ajana, Btihaj. *Governing through Biometrics: The Biopolitics of Identity*. Springer, 2013.

Ajana, Btihaj. 'Personal Metrics: Users' Experiences and Perceptions of Self-Tracking Practices and Data'. *Social Science Information* 59, no. 4 (2020): 654–78. https://doi.org/10.1177/0539018420959522.

Amoore, Louise. 'Biometric Borders: Governing Mobilities in the War on Terror'. *Political Geography* 25, no. 3 (2006): 336–51. https://doi.org/10.1016/j.polgeo.2006.02.001.

Asberg, Cecilia, and Astrida Neimanis. 'Bodies of the Now: Feminist Values in Posthuman Times'. In *Presentación inaugural del congreso: 'Visions of the Nowarts & technology festival'*. Stockholm, 2013.

Bakker, Karen. *Gaia's Web. How Digital Environmentalism Can Combat Climate Change, Restore Biodiversity, Cultivate Empathy, and Regenerate the Earth*. MIT Press, 2014.

Barad, Karen. *Meeting the Universe Halfway*. Duke University Press, 2007.

Benjamin, Ruha. *Race After Technology. Abolitionist Tools for the New Jim Code*. Polity, 2019.

Bost, Suzanne. 'From Race/Sex/etc. to Glucose, Feeding Tube, and Mourning: The Shifting Matter of Chicana Feminism'. In *Material Feminisms*, edited by Stacy Alaimo, and Susan Hekman, 340–72. Indiana University Press, 2008.

Bowker, Geoffrey C., and Susan L. Star. *Sorting Things Out: Classifications and Its Consequences*. The MIT Press, 2000.

Braidotti, Rosi. *Nomadic Theory: The Portable Rosi Braidotti*. Columbia University Press, 2011.

Braidotti, Rosi. *Posthuman Knowledge*. Polity Press Cambridge, 2019.

Butler, Octavia E. *Parable of the Sower*. Four Walls Eight Windows, 1993.

Butler, Octavia E. *Parable of the Talents: A Novel*. Seven Stories Press, 1998.

Cassegård, Carl, and Håkan Thörn. *Post-Apocalyptic Environmentalism: The Green Movement in Times of Catastrophe*. Springer, 2022.

Couldry, Nick, and U. A. Mejias. *The Costs of Connection: How Data Is Colonizing Human Life and Appropriating It for Capitalism*. Stanford University Press, 2020.

Crawford, Kate. *Atlas of AI: Power, Politics, and the Planetary Costs of Artificial Intelligence*. Yale University Press, 2021.

Crossley, Nick, 'Phenomenology and the Body'. In *Routledge Handbook of Body Studies*, edited by Bryan Turner, 130–43. Routledge, 2012.

De Beauvoir, Simone. *What Is Existentialism?* Penguin Books, 2005.

De Beauvoir, Simone. *The Ethics of Ambiguity*. Open Road Integrated Media, 2018.

Descartes, René. *Meditations on First Philosophy: A Biliungual Edition*. University of Notre Dame Press, 1990.

Escobar, Arturo. *Designs for the Pluriverse: Radical Interdependence, Autonomy, and the Making of Worlds*. Duke University Press, 2018.

Ess, Charles M. 'Virtues, Robots, and Good Lives: Who Cares?'. In *Social Robotics and the Good Life: The Normative Side of Forming Emotional Bonds With Robots*, edited by Janina Loh, 25–54. Transcript Verlag, 2023.
Fischer-Lichte, Ericka. *Ästhetik des Performativen*. Suhrkamp Verlag, 2012.
Fry, Tony. *Becoming Human by Design*. A&C Black, 2012.
Fuchs, Thomas. *Verteidigung des Menschen: Grundfragen einer verkörperten Anthropologie*. Suhrkamp Verlag, 2020.
Gates, Kelly A. *Our Biometric Future*. New York University Press, 2011.
Haraway, Donna J. 'SF: Science Fiction, Speculative Fabulation, String Figures, So Far'. *Ada: A Journal of Gender, New Media, and Technology* (2013). https://scholarsbank.uoregon.edu/xmlui/bitstream/handle/1794/26308/ada03-sfsci-har-2013.pdf?sequence=1.
Haraway, Donna J. *Simians, Cyborgs, and Women: The Reinvention of Nature*. Routledge, 2013.
Haraway, Donna J. *Staying with the Trouble: Making Kin in the Chthulucene*. Duke University Press, 2016.
Hook, Kristina. *Designing with the Body: Somaesthetic Interaction Design*. MIT Press, 2018.
Howell, Noura. *Emotion AI and Fabulation: Seeking Biopolitical Futures of Respectful Care*. WASP-HS Crafting Livable Futures. Umeå University, 2023.
Jaspers, Karl. *The Origin and Goal of History*. Routledge, 2011.
Kimmerer, Robin W. *Braiding Sweetgrass: Indigenous Wisdom, Scientific Knowledge and the Teachings of Plants*. Milkweed Editions, 2013.
Lagerkvist, Amanda. 'Existential Media: Toward a Theorization of Digital Thrownness'. *New Media & Society* 19, no. 1 (2017): 96–110. https://doi.org/10.1177/1461444816649921.
Lagerkvist, Amanda. 'Digital Limit Situations: Anticipatory Media Beyond "The New AI Era"'. *Journal of Digital Social Research (JDSR)* 2, no. 3 (2020): 16–41.
Lagerkvist, Amanda. *Existential Media: A Media Theory of the Limit Situation*. Oxford University Press US, 2022.
Lagerkvist, Amanda, Matilda Tudor, Jacek Smolicki, et al. 'Body Stakes: An Existential Ethics of Care in Living with Biometrics and AI'. *AI & Society: The Journal of Human-Centred Systems and Machine Intelligence* 39, no. 1 (2020). https://doi.org/10.1007/s00146-022-01550-8.
Le Guin, Ursula K. *The Carrier Bag Theory of Fiction*. St. Martin's Press, 1988.
Lupton, Deborah. *The Quantified Self*. Polity Press, 2016.
Martin, Aryn, Natasha Myers, and Ana Viseu. 'The Politics of Care in Technoscience'. *Social Studies of Science* 45, no. 5 (2015): 625–41.
Mayer-Schönberger, Victor, and Kenneth Cukier. *Big Data: A Revolution that will Transform How We Live, Work, and Think*. John Murray Publishers, 2013.
Merleau-Ponty, Maurice. *Phenomenology of Perception*. Routledge, 2013.
Nancy, Jean-Luc. *Being Singular Plural*. Stanford University Press, 2000.

Peters, John D. 'What Is Not a Medium'. *Communication +1* 9, no. 1 (2022). doi: https://doi.org/10.7275/epdv-p307.
Petersén, Moa. *The Swedish Microchipping Phenomenon*. Emerald Publishing Limited, 2019.
Puig de la Bellacasa, Maria. *Matters of Care: Speculative Ethics in More than Human Worlds*. University of Minnesota Press, 2017.
Rhee, Jennifer. *The Robotic Imaginary: The Human and the Price of Dehumanized Labor*. University of Minnesota Press, 2018.
Salami, Minna. *Sensuous Knowledge: A Black Feminist Approach for Everyone*. Bloomsbury Publishing, 2020.
Tsing, Anna L. *The Mushroom at the End of the World: On the Possibility of Life in Capitalist Ruins*. Princeton University Press, 2015.
Tudor, Matilda. 'Queering Digital Media Spatiality: A Phenomenology of Bodies Being Stopped'. *Feminist Media Studies* 23, no. 3 (2021): 1–16. https://doi.org/10.1080/14680777.2021.1980080.
Turner, Bryan S. *The Body and Society: Explorations in Social Theory*. Sage, 2008.
van Dijck, Jose. 'Datafication, Dataism and Dataveillance: Big Data between Scientific Paradigm and Ideology'. *Surveillance & Society* 12, no. 2 (2014): 197–208. https://doi.org/10.24908/ss.v12i2.4776.
Zuboff, Shoshanna. *The Age of Surveillance Capitalism: The Fight for a Human Future at the New Frontier of Power*. Profile Books, 2019.